

ALEKSANDAR DUBLJEVIC

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EDUCATION

University of Waterloo & Wilfrid Laurier University

Bachelor of Computer Science and Bachelor of Business Administration Double Degree

Awards: Pearl Sullivan Hack the North Scholarship | Relevant Coursework: Data Structures and Algorithms

Expected May 2028

Waterloo, ON

EXPERIENCE

AI Engineer

Jan 2026 – Apr 2026; Returning Fall 2026

RBC Capital Markets

Toronto, ON; New York, NY

- Architected a text-to-SQL **MCP** server enabling **35+ traders** to query **410M+ rows** of live and historical trade data in natural language, with insights supporting **\$118M+** in trades during its first week in production
- Built **MLflow** evaluation pipelines for an LLM news retrieval system used by **650+ employees**, enabling benchmarking that identified failure modes and validated fixes that improved accuracy from **35% to 95%**
- Partnered with traders, engineers, database owners, and compliance to deploy MCP services with **Docker, Helm, and Kubernetes**, adding **RBAC and audit logging** for secure use across multiple desks
- Developed AI agents using the **Claude Agents SDK**, creating a shared library of reusable skills and sub-agents that automated multi-step workflows, increasing median monthly AI usage from **7 to 15 days per user**

Cloud Engineer

Jan 2025 – Apr 2025

Canadian Imperial Bank of Commerce

Toronto, ON

- Deployed and tested **Azure** resource configurations (Databricks, VMs, Storage) for enterprise applications across dev/test/prod environments, completing **63 Jira tickets** with zero post-deployment rollback incidents as part of an Agile Scrum team
- Resolved a **SEV2** outage impacting critical banking services by tracing network flows with **Azure Network Watcher** to identify misconfigured firewall rules, collaborating with senior engineers to restore connectivity for thousands of users within 2 hours
- Automated Azure resource deployments by executing **Terraform** scripts through **Azure DevOps CI/CD** pipelines, reducing manual provisioning time by **42%** while ensuring version-controlled, consistent infrastructure across environments

Software Engineer - Dashboard Lead

Sep 2023 – Dec 2025

University of Waterloo Formula Electric

Waterloo, ON

- Led the development of an internal **Grafana** dashboard for live telemetry, enabling team members to monitor real-time performance from over **150 data sources**, eliminating the need to restart the vehicle to view sensor data
- Built a **Debian Linux** heads-up display, implementing real-time CAN bus message processing on an embedded system
- Built HIL safety tests for precharge and discharge control logic, validating high-voltage embedded systems

PROJECTS

LLM RL Training (Qwen3-4B) 🦾 | Python, Tinker, vLLM, Hugging Face

- Trained a 4B parameter Qwen model to **99%+ solve accuracy** on NYT Connections puzzles via supervised fine-tuning (LoRA) followed by GRPO-based **reinforcement learning**
- Designed a **verifiable reward scheme** around objectively checkable groupings, incentivizing correctness and token efficiency
- Deployed the model locally using **vLLM**, constraining context size and leveraging prefix caching to reduce redundant computation during multi-turn inference

Dime Defender 🦾 | Hack the North 2024 Finalist — AWS, ElevenLabs, Voiceflow, OpenAI API

- Top 12** among **800+ participants** at Hack the North 2024 for developing an AI-powered Chrome extension that generated contextual spending interventions during e-commerce checkout
- Architected a scalable **AWS** serverless backend using Lambda and API Gateway that supported over **75 concurrent users**, achieving **sub-500ms** response times through API request optimizations including pagination and payload compression

MpoxMap 🦾 | Ignition Hacks 2024 Winner — TypeScript, React, Next.js, HTML/CSS

- Developed a Next.js and React web app that visualizes real-time and historical Mpox outbreaks on an interactive 3D map
- Implemented API caching to reduce redundant calls to TheNewsAPI and Mapbox API, lowering page-load latency and API costs while improving application responsiveness

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, Shell Script, Go, Rust

AI/ML & Frameworks: MLflow, Hugging Face, vLLM, LangChain, Llamaindex, Claude Agents SDK, React, Node.js, Next.js

Cloud & Developer Tools: Azure, AWS, Kubernetes, Docker, Terraform, GitHub Actions, Git, GitHub, Claude Code, Codex